Entropy-Complete Plaquette Activities

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Abstract

We audit the corrected collar charge against radial integration entropy. The bare action criterion closes at the current coherence radius, while the entropy-complete Gaussian plaquette activity does not.

For a three-component Gaussian curvature with $s = \rho^2/t$, the quadratic Chernoff rate is

$$J_3(s) = \frac{1}{2}\left(s - 3 - 3\log(s/3)\right), \qquad s > 3.$$

The bare action exponent is $s/2$; their difference is the radial entropy cost.

Theorem 1. *For dependent projected Gaussian curvature, every connected k-plaquette event is bounded by $e^{-kJ_3(s)}$ without assuming independence. With Bianchi degree D, the rooted entropy-complete series converges only if*

$$e^{\alpha} D^2 e^{-J_3(s)} < 1.$$

Its size-one rooted free energy is $J_3(s) - \alpha + \log(1 - e^{\alpha} D^2 e^{-J_3(s)})$.

Proof. The joint quadratic exponential moment yields the first factor. Multiplication by the connected-set bound $D^{2(k-1)}$ and the state weight $e^{\alpha k}$ gives a geometric series. □

At the current coherent eta ceiling and $D = 20$, the bare action series is subcritical but the entropy-complete series is supercritical. The executable inversion reports the larger curvature required for convergence and for free energy 6.71292101139.

Remark 1. *This falsifies an action-only promotion. It is a Gaussian reference theorem; nonlinear normalized comparison remains open.*